

Restaurant Reservation and Management System Project Idea

1. Project Objectives

The primary objectives for developing a Restaurant Reservation and Management System are to:

- **Streamline Reservation Processes:** Automate the booking of tables, reducing manual errors and improving efficiency for both customers and staff.
- **Easy Online Delivery:** Allows customers to place orders online for delivery.
- **Enhance Customer Experience:** Provide a user-friendly interface for customers to easily make, modify, or cancel reservations at their convenience.
- **Optimize Table Management:** Enable restaurants to effectively manage table availability, seating arrangements, and peak hour demands.
- **Improve Operational Efficiency:** Offer tools for staff to manage reservations, customer data, and restaurant capacity, leading to better service delivery.

2. Project Scope

This project will focus on developing a full-stack web application using the MERN (MongoDB, Express.js, React.js, Node.js) stack. The key features to be included are:

- **Customer-facing portal:**
 - User registration and login
 - Browse available restaurants and table times
 - Make, view, modify, and cancel reservations
 - Receive reservation notifications
- **Restaurant staff-facing portal (Manager and Admin):**
 - Admin login and user management
 - View and manage all reservations
 - Table management (assigning tables, marking tables as occupied/available)
 - Customer database management
 - Status updates on reservations and customer data
- **API Development:** Secure and efficient APIs for communication between the front-end and back-end.

- **Database Management:** MongoDB for flexible and scalable data storage.

3. Expected Outcomes

Upon successful completion, the Restaurant Reservation and Management System is expected to deliver the following outcomes:

- **Increased Reservation Accuracy:** Significantly reduce errors associated with manual booking.
- **Higher Customer Satisfaction:** Offer a convenient and reliable reservation experience.
- **Optimized Table Utilization:** Maximized seating capacity and reduced wait times.
- **Reduced Administrative Burden:** Free up staff time currently spent on managing reservations manually.

4. Justification of Chosen Idea

The Restaurant Reservation and Management System was chosen because it addresses real-world challenges faced by restaurants in managing table bookings, orders, and customer feedback efficiently. Many restaurants still rely on manual or phone-based systems, which often lead to errors, overbooking, and customer dissatisfaction. A web-based solution offers convenience for both customers and administrators

Customers can easily reserve tables and order food online, while admins can manage bookings, update menus, and view feedback in real time. This project also provides an opportunity to apply key software engineering concepts such as database management, user authentication, and responsive web design in a practical, impactful way.

Restaurant Management System User Stories & Architecture

Software Testing Project Document

User Stories:

- **FR1 Customer Table Reservation**
 - As a customer, I want to view real-time available tables based on party size, date, and time, so that I can make an informed decision when creating a new reservation.
 - As a customer, I want to be able to order food while reserving a table (optional), so that I can streamline my dining experience.
- **FR2 Reservation Management**
 - As a customer, I want to be able to modify or cancel an existing reservation, so that I have flexibility with my plans.
 - As an admin user, I want to be able to modify or cancel an existing reservation, so that I can manage table availability efficiently.
 - As the system, I want to update table availability instantly upon modification or cancellation, so that real-time accuracy is maintained.
- **FR3 Menu/Dish Management**
 - As an admin user, I want to be able to create, read, update, and delete menu items, including details like price, description, and images, so that I can keep the restaurant's menu current and accurate.
- **FR4 Online Food Ordering**
 - As a registered customer, I want to view the menu, add items to a cart, and place a take-away or delivery order, so that I can conveniently order food online.
- **FR5 Role-Based Authentication**
 - As a user, I want to securely log in and log out, so that my account is protected.
 - As an authenticated customer, I want to be able to place orders and make reservations, so that I can utilize the system's features.
 - As an admin, I want to be able to access the administrator panel, so that I can manage the restaurant operations.
- **FR6 Order Status Tracking & Reporting**
 - As a manager, I want to view, accept, and update the status of all customer orders (e.g., Pending, In Progress, Completed), so that I can manage order fulfillment effectively.
 - As an admin, I want to generate reports (e.g., daily revenue, popular items), so that I can gain insights into restaurant performance.

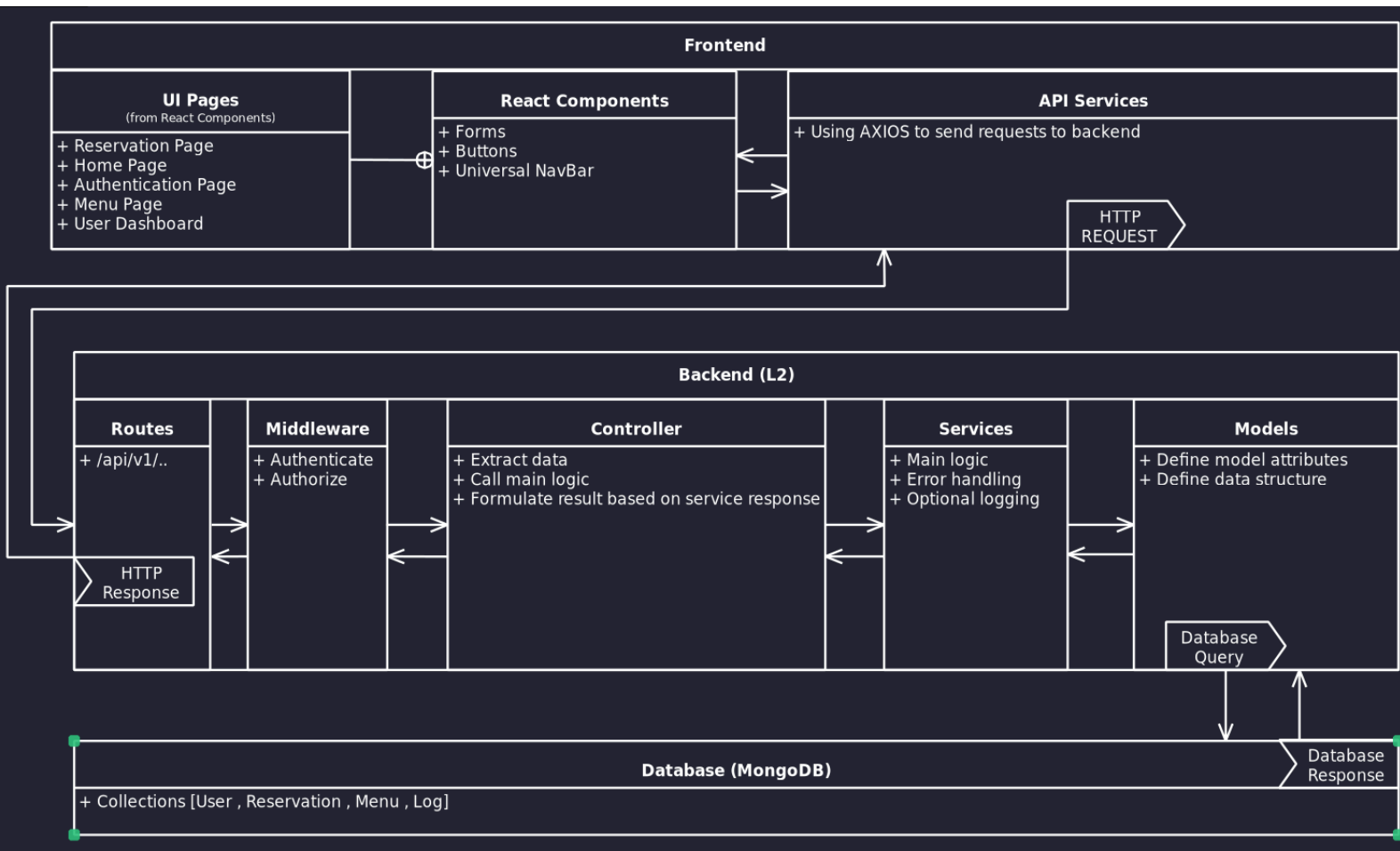
- **FR7 Customer Support**
 - As a customer, I want to open a ticket for service inquiries without logging in, so that I can easily get support.
- **FR8 Allow account deletion & modification**
 - As a user, I want to be able to delete my account, so that I have control over my personal data.
 - As a user, I want to be able to modify my account details, so that I can keep my information up-to-date.
- **FR9 Restaurant Manager**
 - As a restaurant manager, I want to update order status for customers and view them for fulfillment from the management page, so that I can efficiently manage customer orders.
- **FR10 Customer Feedback**
 - As a customer, I want to leave reviews, so that I can share my dining experience.
 - As an admin/manager, I want to view customer reviews, so that I can understand customer satisfaction and identify areas for improvement.
- **FR11 Notification System**
 - As a staff member, I want to receive alerts for new reservations, so that I can be aware of upcoming bookings.
 - As a customer, I want to receive an email with my reserved table and time, so that I have confirmation of my booking.
- **FR12 User Wallet**
 - As a user, I want to have a personal wallet, so that I can receive refunds issued by the admin.

System Architecture:

The system will be layer architected for separation of concerns, easy maintainability and easier testing.

The layers will be React (Frontend), Express & [Node.js](#) (backend) and MongoDB for Database

Architecture Diagram:



Bonus:

Use Case Diagram

